

Context: thinking outside the (bounding) box



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16-721: Learning-Based Methods in Vision
A. Efros, CMU, Spring 2009

Eye of the Beholder



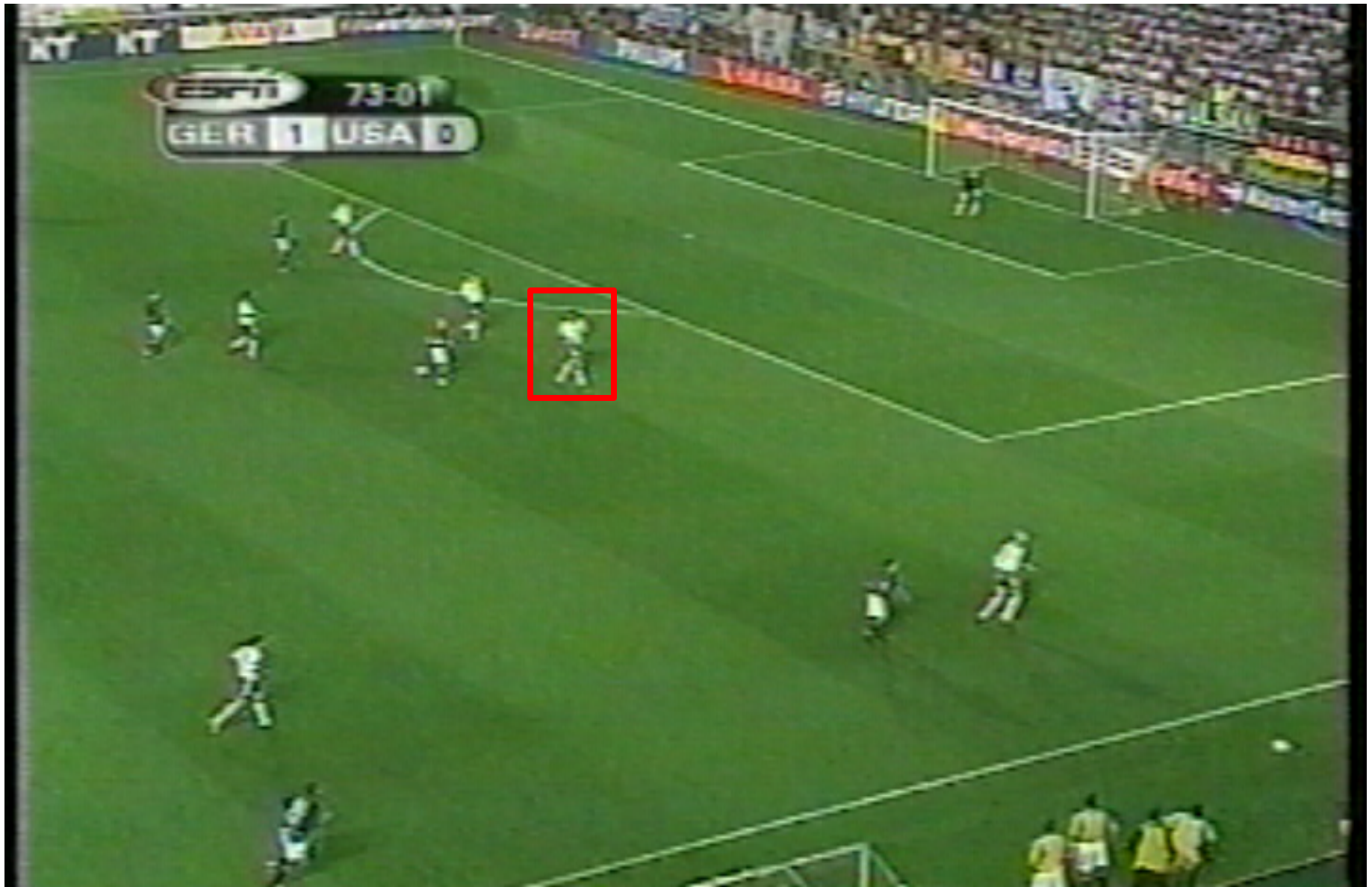
**Claude
Monet**
Gare St.Lazare
Paris, 1877

Eye of the Beholder



where did it go?

Seeing less than you think...



Seeing less than you think...



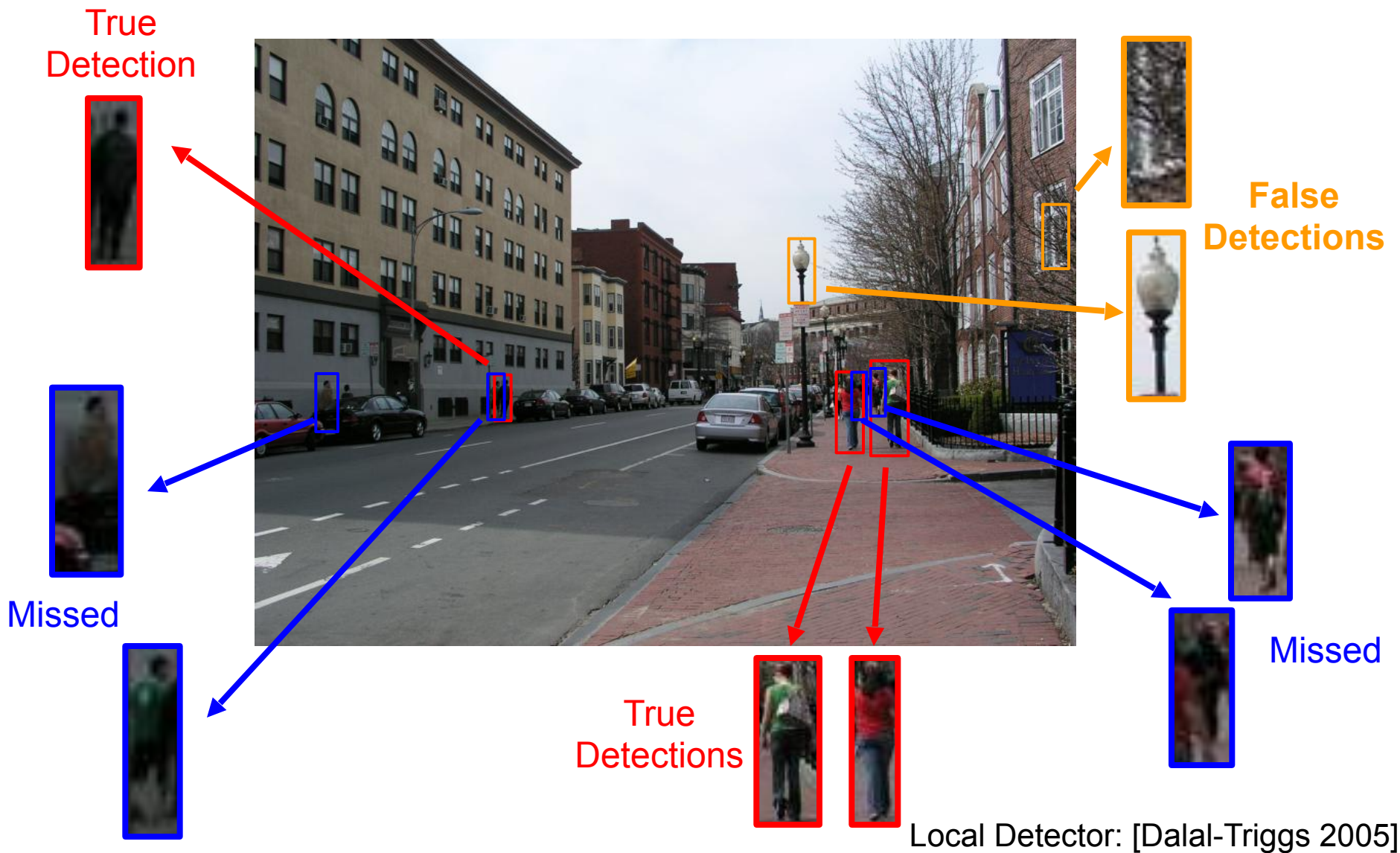
“The Miserable Life of a Person Detector”



What the Detector Sees



What the Detector Does



with hundreds of categories...



If we have 1000 categories (detectors), and each detector produces 1 FP every 10 images, we will have 100 false alarms per image... pretty much garbage...

Context to the rescue!

We know there is a keyboard present in this scene even if we cannot see it clearly.

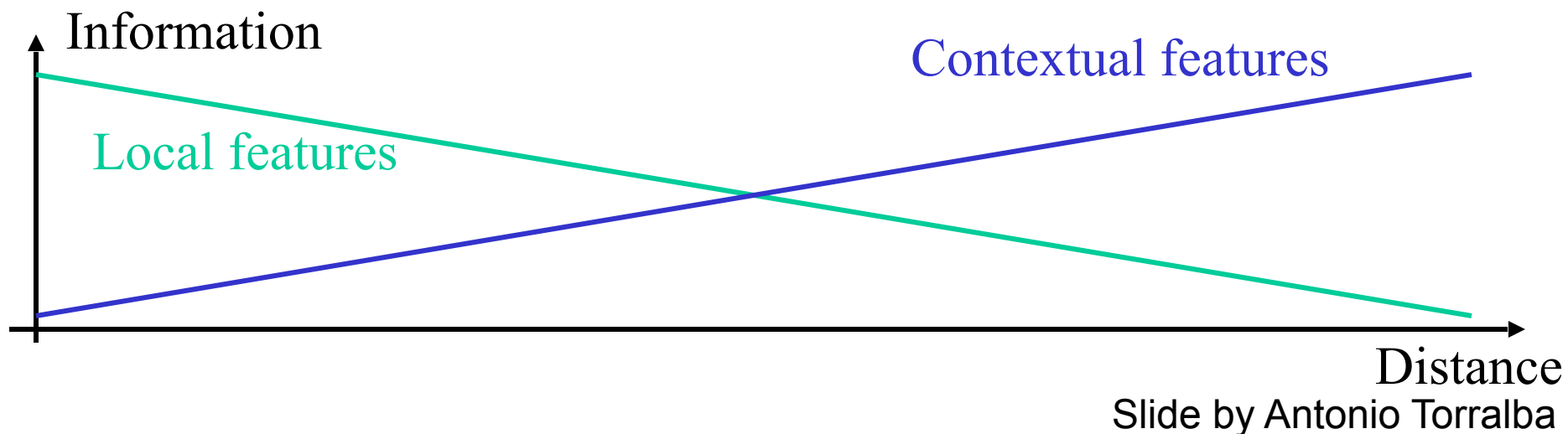


We know there is no keyboard present in this scene



... even if there is one indeed.
Slide by Antonio Torralba

When is context helpful?



Humans can get a lot from context...



What is Rob Fergus doing?

Clip by Antonio Torralba

even when it's wrong!



What is Rob Fergus *really* doing???

Clip by Antonio Torralba

Is it just for small / blurry things?

A B C

Is it just for small / blurry things?

12

13

14

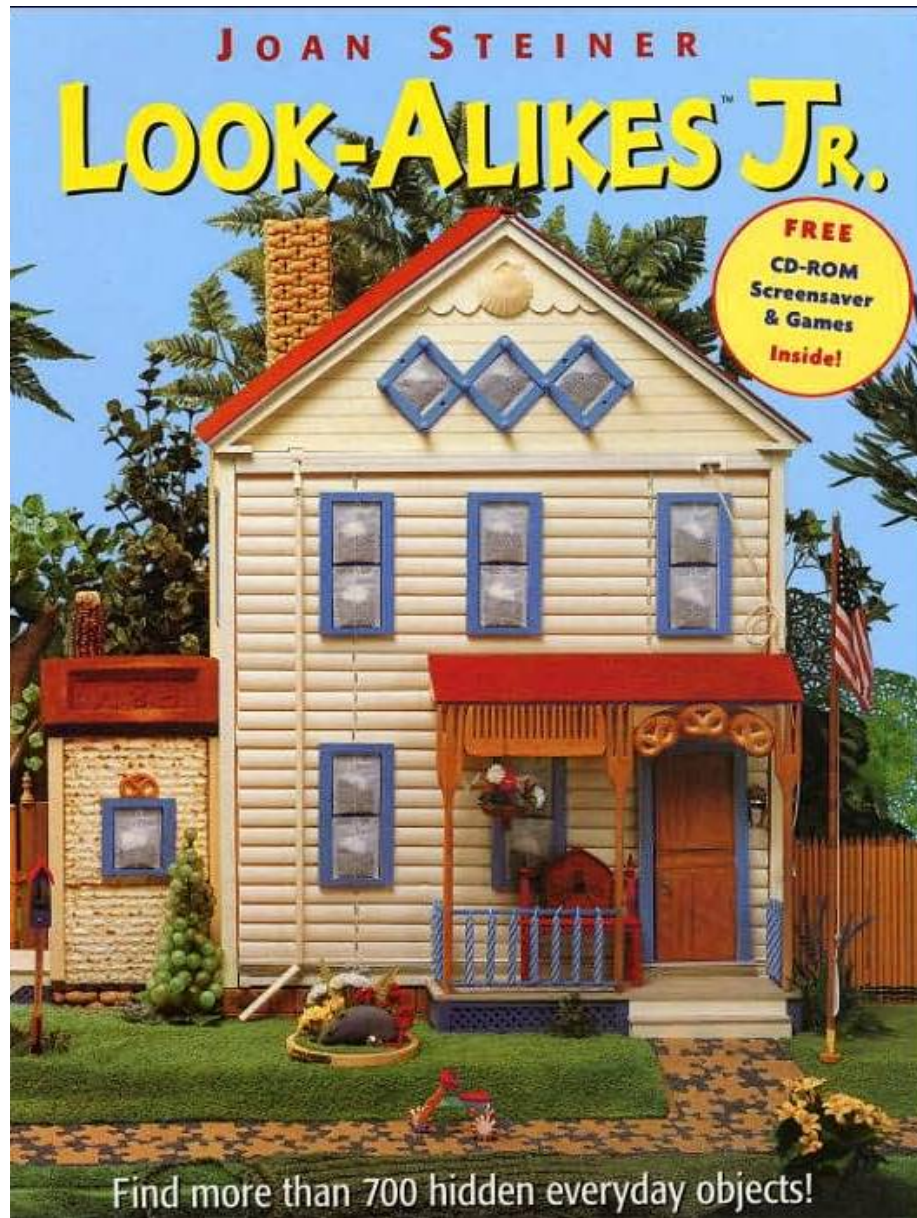
Is it just for small / blurry things?

A B C

12
13
14

12
A B C
14

Context is hard to fight!



Thanks to Paul Viola
for showing me these

more “Look-alikes”



Don't even need to see the object



Don't even need to see the object

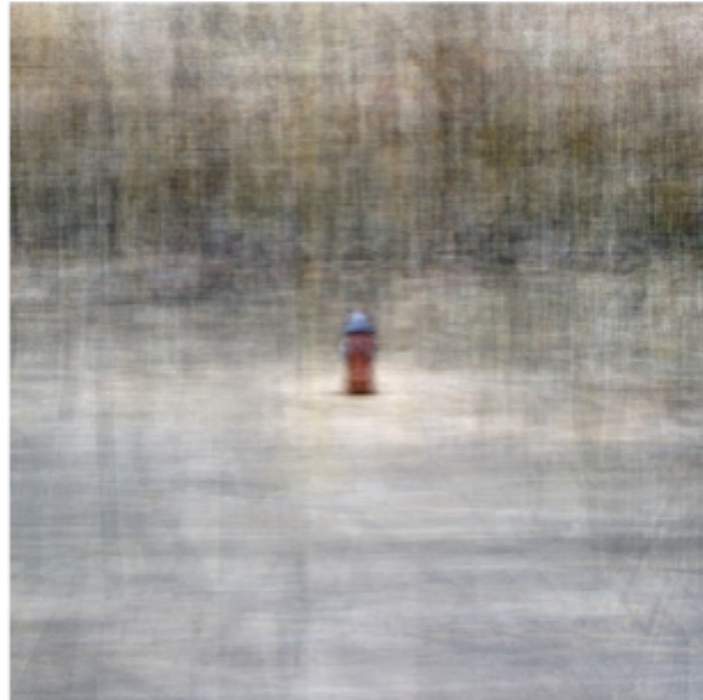


Slide by Antonio Torralba

But object can say a lot about the scene



The influence of an object extends beyond its physical boundaries



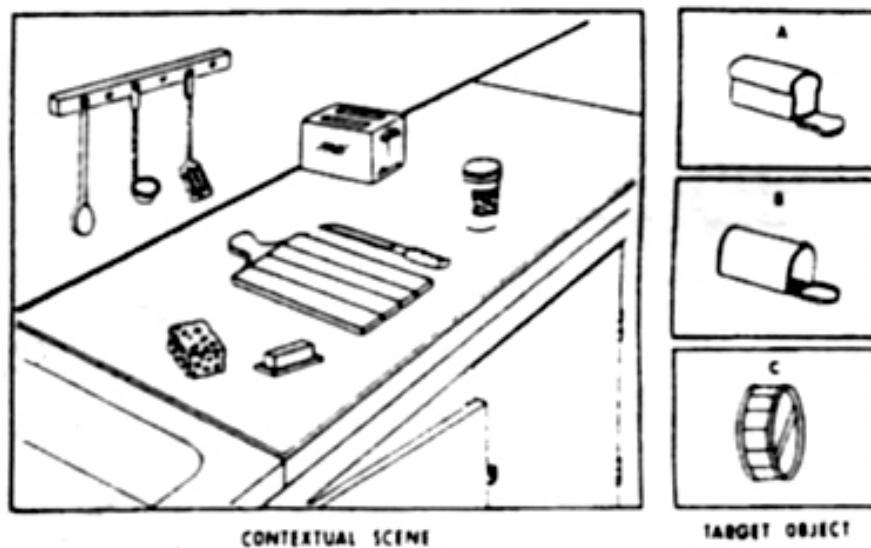
TRENDS in Cognitive Sciences

Context in Human Perception

Palmer 1975

Scene preceded object to identify.

Better identification when preceded by a semantically consistent scene.



Objects seen for 20, 40, 60 or 120 ms. Slide by Antonio Torralba

Palmer

Scenes shown ahead of time for 2 s.

More accurate recognition of consistent objects than inconsistent objects.

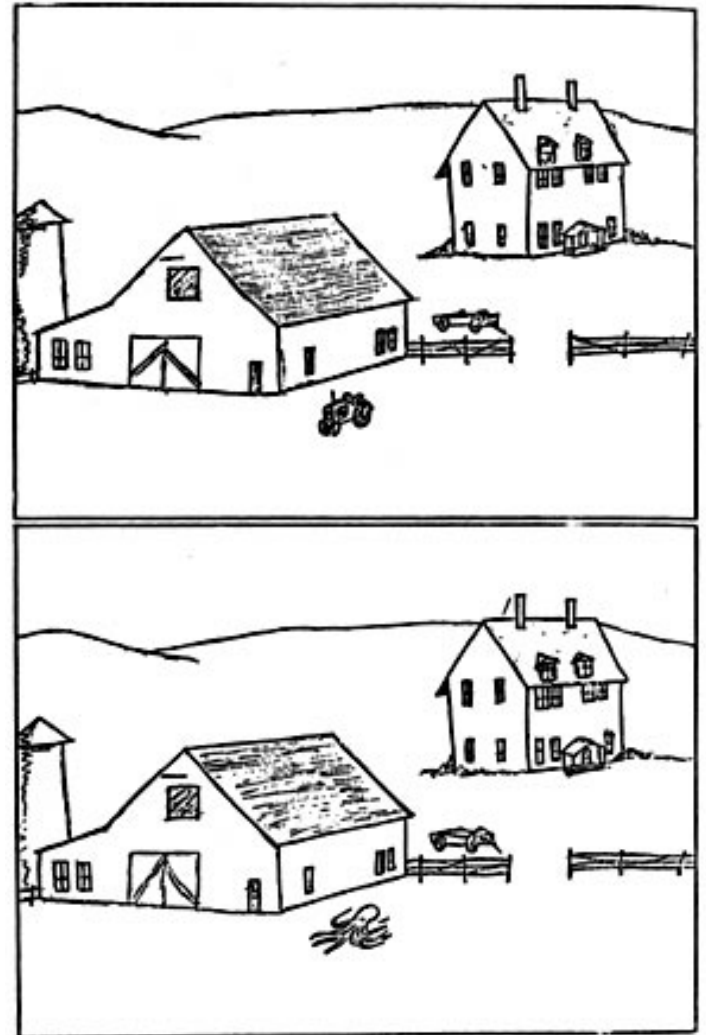
Similar looking objects were misnamed, showing a bias effect.

Loftus & Mackworth

Inconsistent objects fixated earlier and longer.

Suggested additional processing of objects out of context.

Similar results found by Friedman (1979).



Hock et al.



TYPE I



TYPE II



TYPE III



TYPE IV

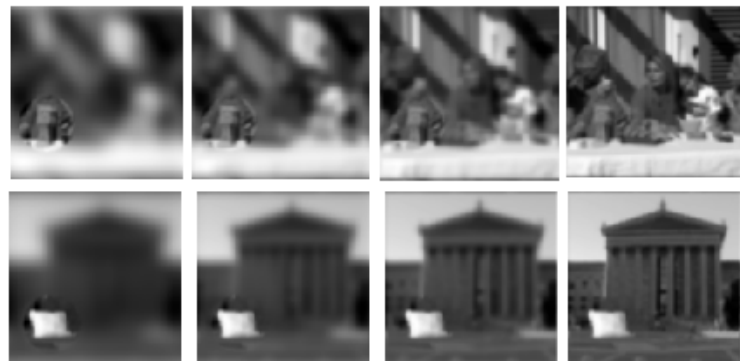
Object priming



Increasing contextual information

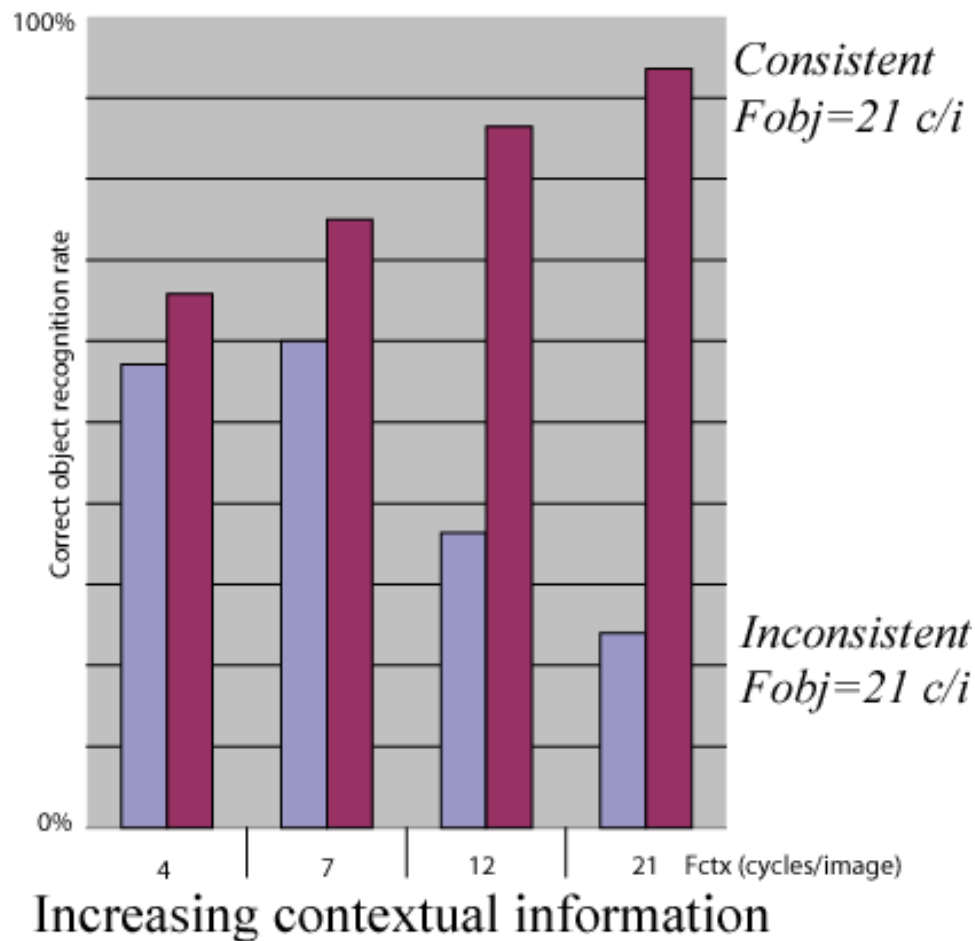
Object priming

Inconsistent objects



$F_{ctx}=4$ 7 12 21

Consistent objects

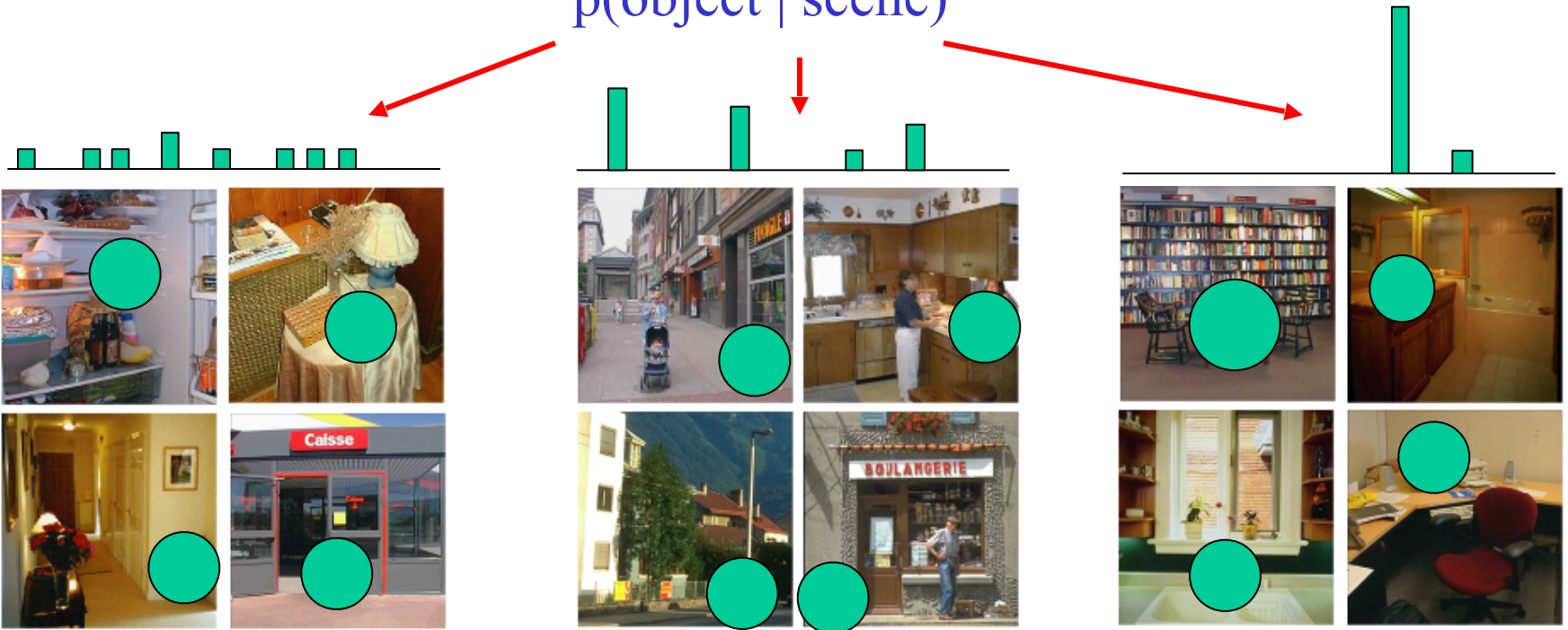


Object priming



→ Car, pedestrian, mailbox, ...

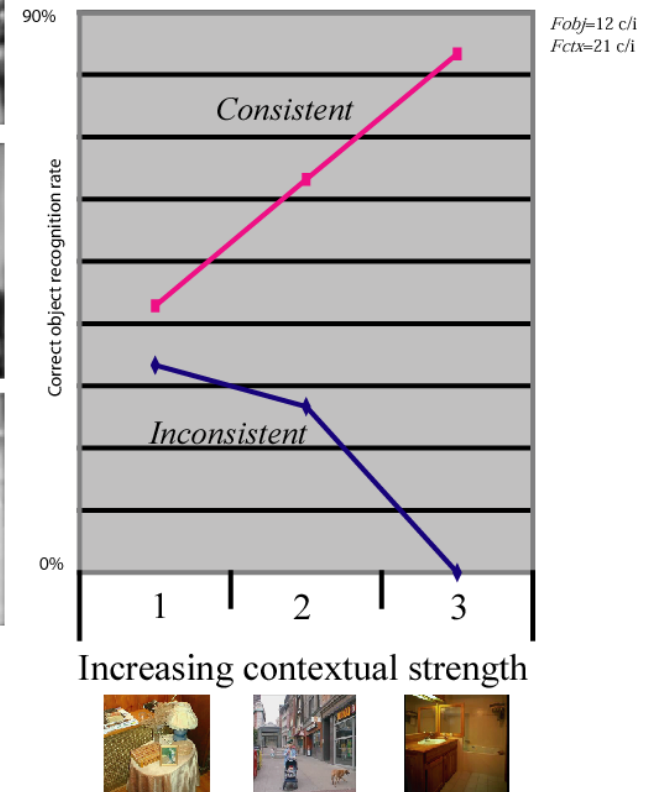
$p(\text{object} \mid \text{scene})$



Low rigidity

Strong rigidity

Object priming



a



b



c

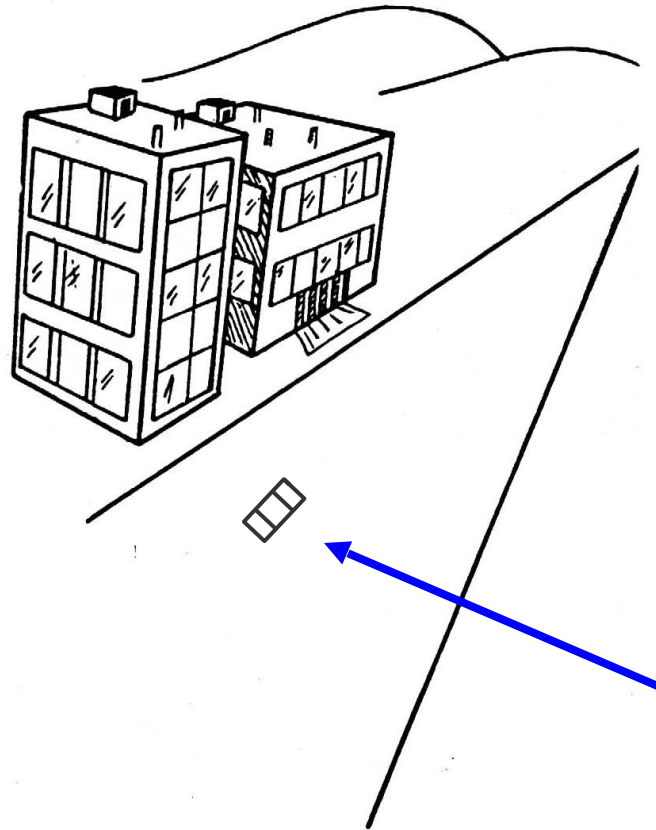


Examples of consistent scenes (a), inconsistent scenes (b), and isolated objects and backgrounds (c); from Davenport & Potter, 2004

Why context is important?

- Typical answer: to “guess” small / blurry objects based on a prior
 - most current vision systems

So, you think it's so simple?

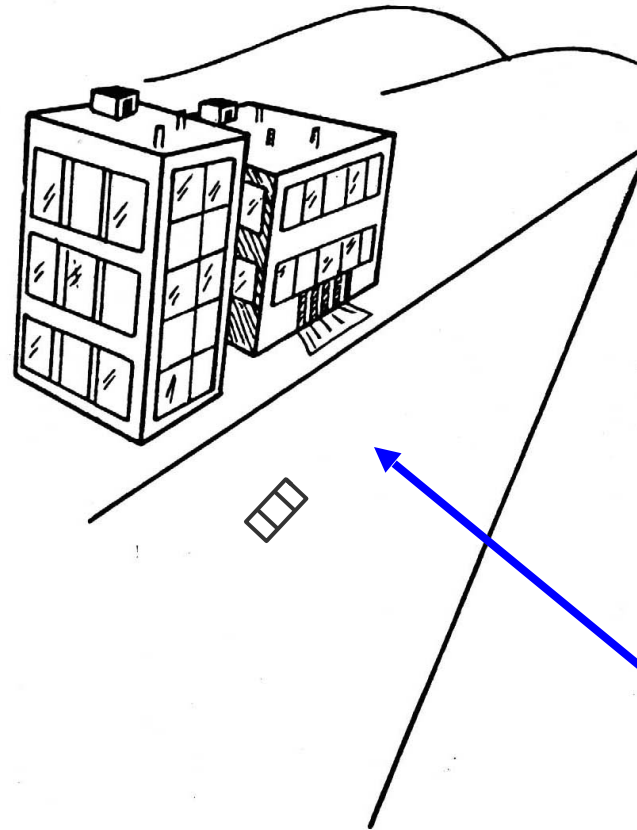


What is this?

Why is this a car?

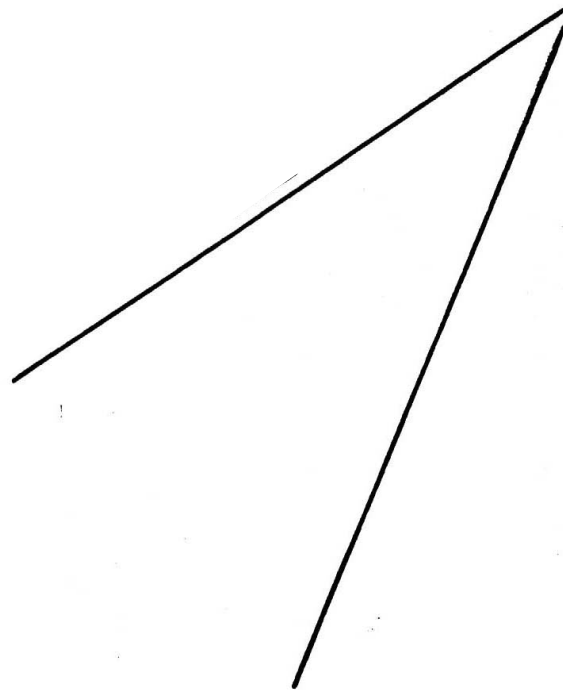


...because it's on the road!

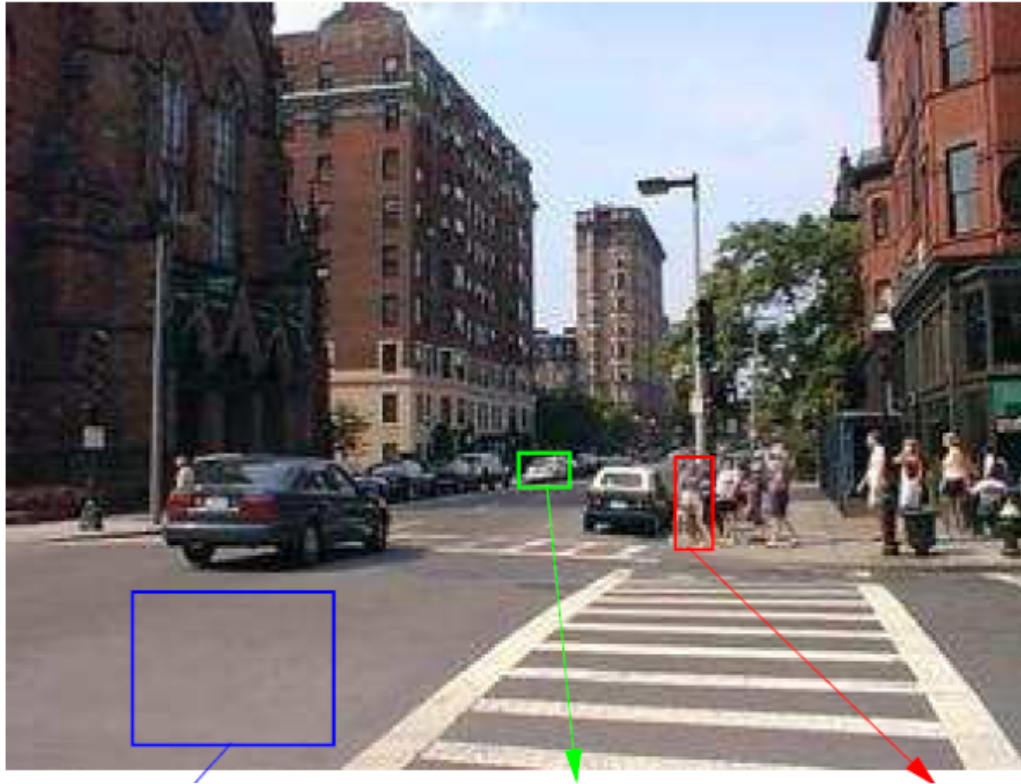


Why is this road?

Why is this a road?



Same problem in real scenes



Why context is important?

- Typical answer: to “guess” small / blurry objects based on a prior
 - Most current vision systems
- Deeper answer: to make sense of the visual world
 - much work yet to be done!

Why context is important?

- To resolve ambiguity
 - Even high-res objects can be ambiguous
 - e.g. there are more people than faces in the world!
 - There are 30,000+ objects, but only a few dozen can happen within a given scene
- To notice “unusual” things
 - Prevents mental overload
 - Autistic people may lack this ability
- To infer function of unknown object



Sources of Context

Local Pixel Context	window surround, image neighborhoods, object boundary/shape
Scene Gist Context	global image statistics
Geometric Context	3D scene layout, support surface, surface orientations, occlusions, contact points, etc.
Semantic Context	event/activity depicted, scene category, objects present in the scene, keywords
Photogrammetric Context	camera height, orientation, focal length, lens distortion, radiometric response function
Illumination Context	sun direction, sky color, cloud cover, shadow contrast, etc
Weather Context	current/recent precipitation, wind speed/direction, temperature, the season, etc.
Geographic Context	GPS location, terrain type, land use category, elevation, population density, etc.
Temporal Context	nearby frames (if video), temporally proximal images, videos of similar scenes
Cultural Context	photographer bias, dataset selection bias, visual clichés, etc

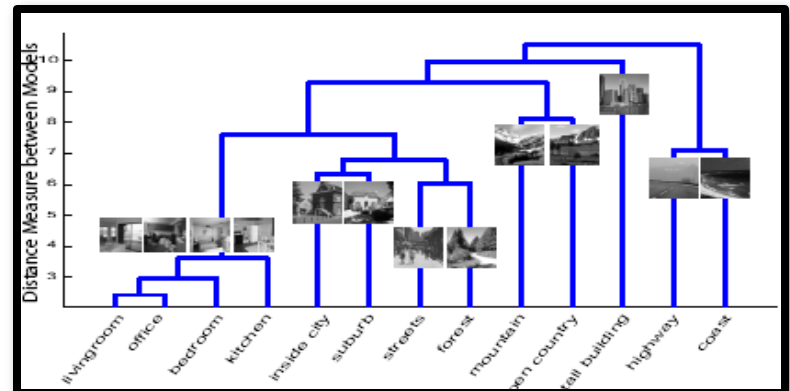
Table 1. Taxonomy of various sources of contextual information.

from Divvala et al, CVPR 2009

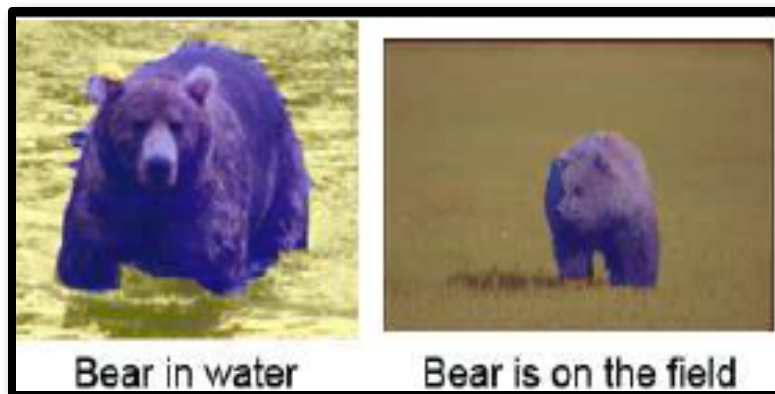
Semantic Context



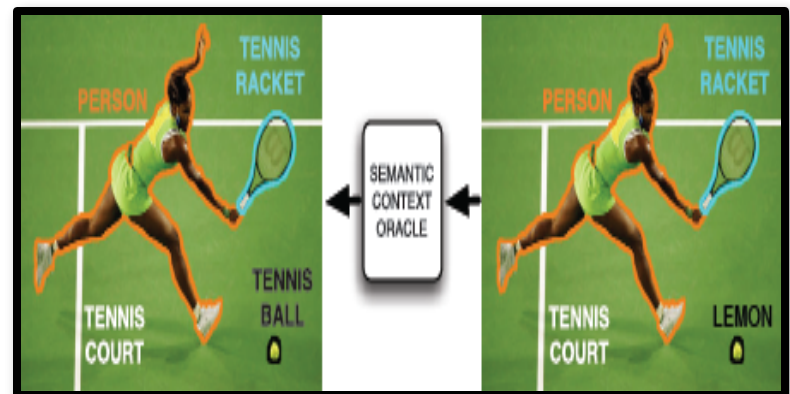
Berg et al. 2004



Fei-Fei and Perona 2005

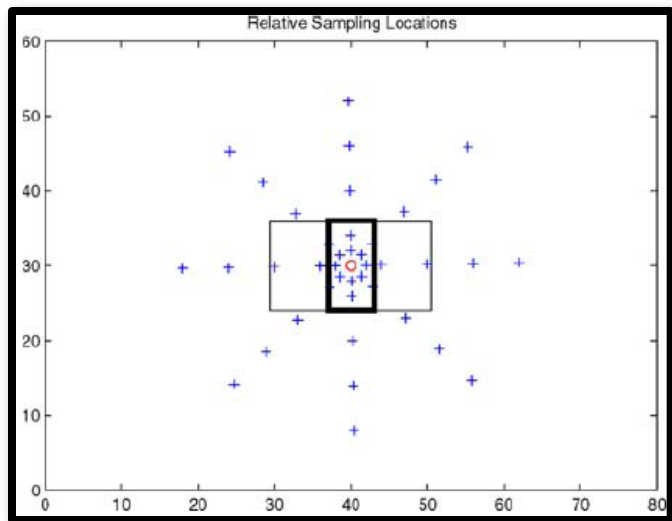


Gupta & Davis 2008

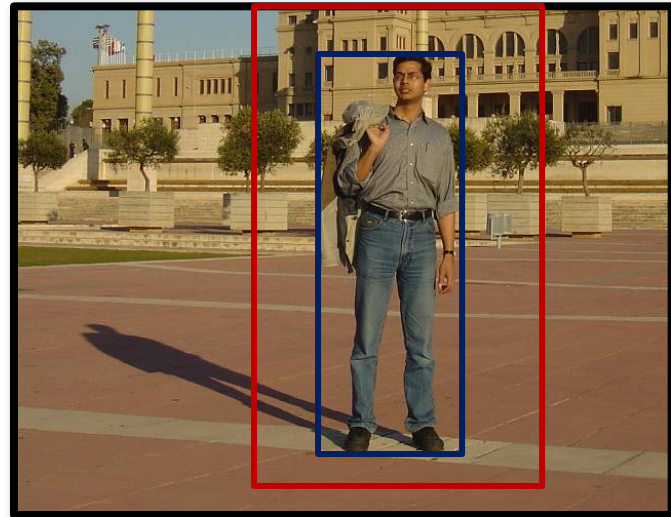


Rabinovich et al. 2007

Local Pixel Context

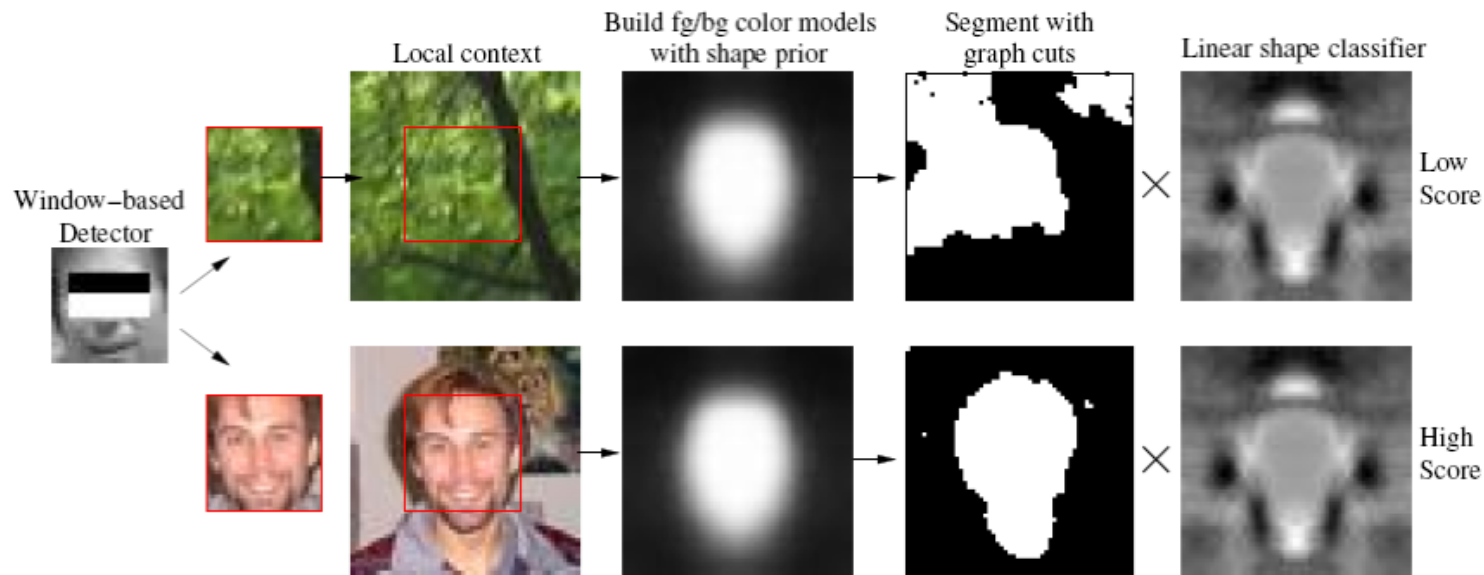


Wolf & Bileschi 2006



Dalal & Triggs 2005

Segmentation is a type of pixel context

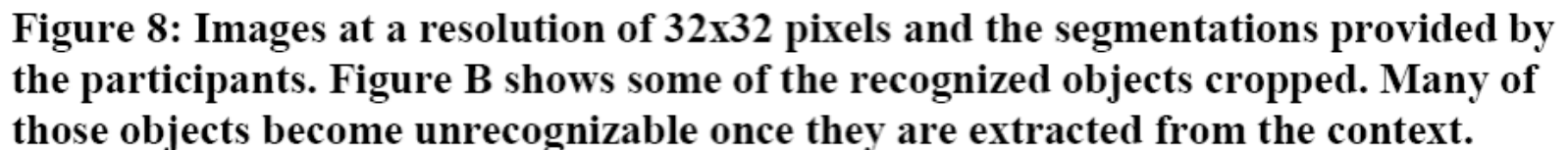


Ramanan, 2007



More generally...

Global (low-dimensional) scene statistics



Geometric Context



TYPE I



TYPE II



TYPE III



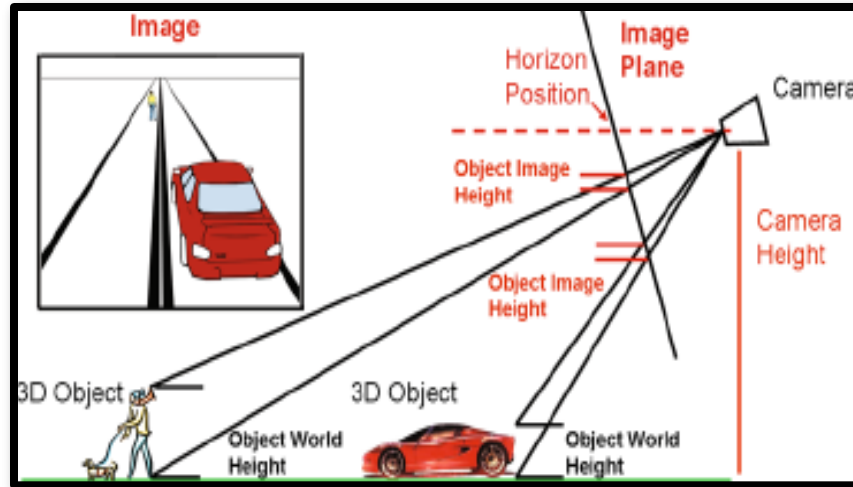
TYPE IV

Hock et al, Biederman

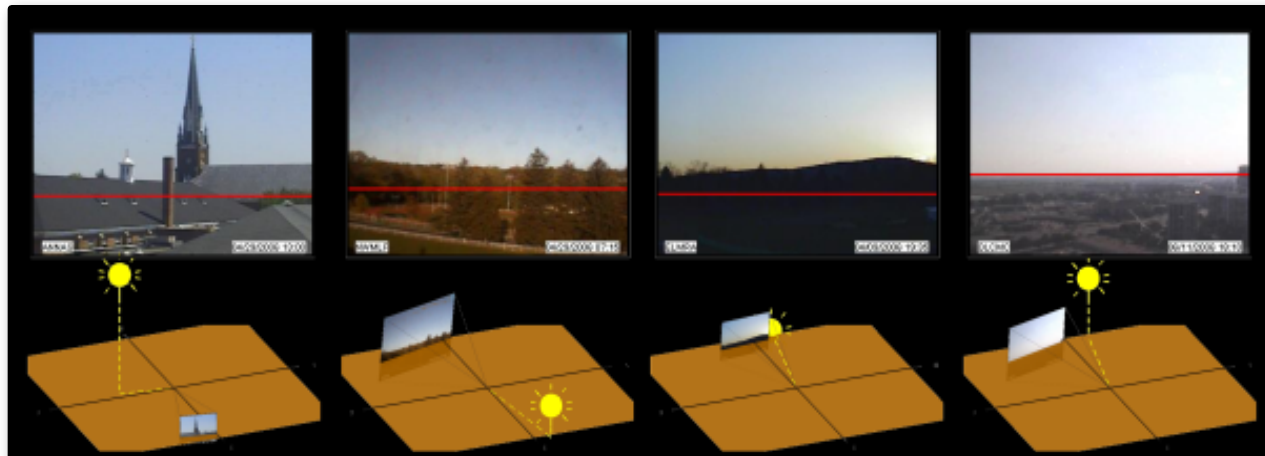


Geometric Context [Hoiem *et al.* '2005]

Photogrammetric Context



Hoiem et al 2006



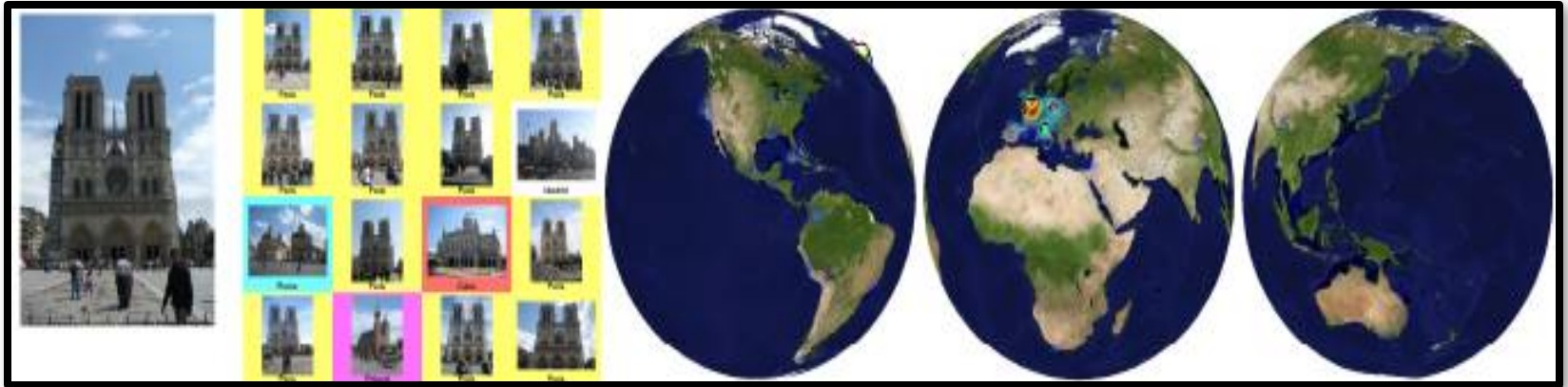
Lalonde et al. 2008

Illumination and Weather Context



Illumination context (Lalonde et al)

Geographic Context

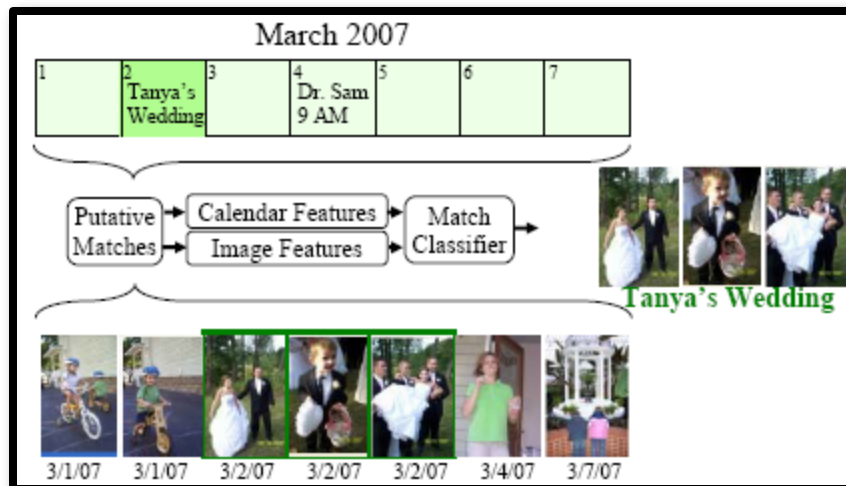


Hays & Efros 2008

Temporal Context



Liu et al. 2008



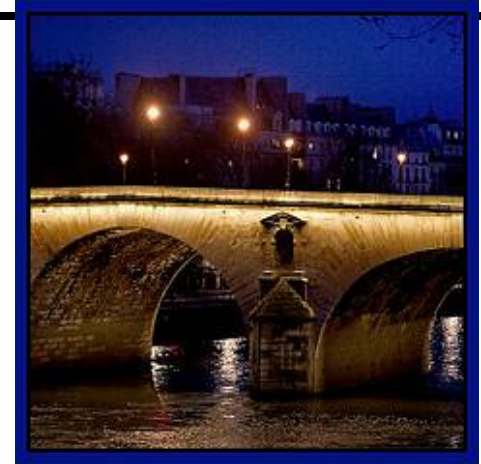
Gallagher et al 2008

Cultural Context

Photographer Bias

Society Bias

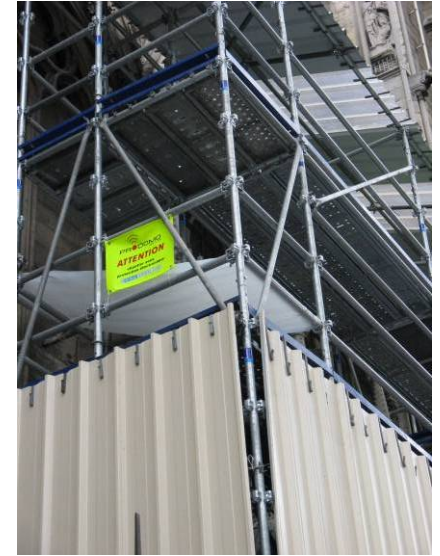
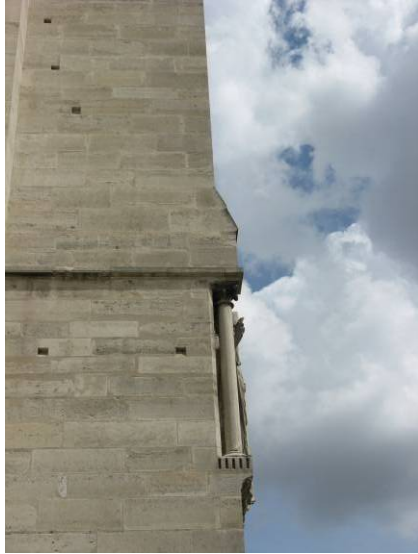
Flickr Paris



“uniformly sampled” Paris

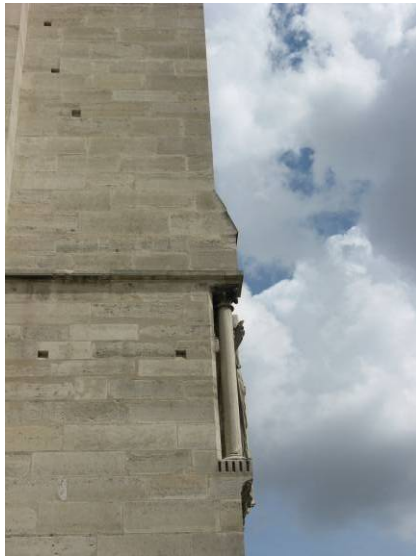


...or Notre Dame



Why Photographers are Biased?

People want their pictures to be recognizable and/or interesting



vs.



Why Photographers are Biased?

People follow photographic conventions



VS.



Simon & Seitz 2008

Cultural context



Lisa and Mildred. Which is which?

Cultural Context

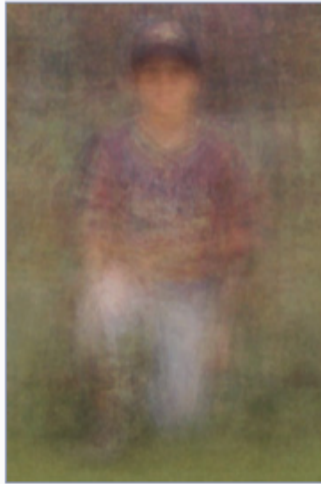


Gallagher et al 2008

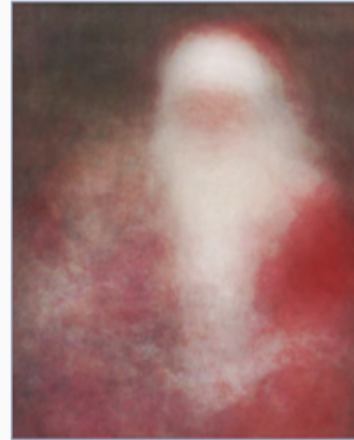


Gallagher et al, CVPR 2009

“100 Special Moments” by Jason Salavon



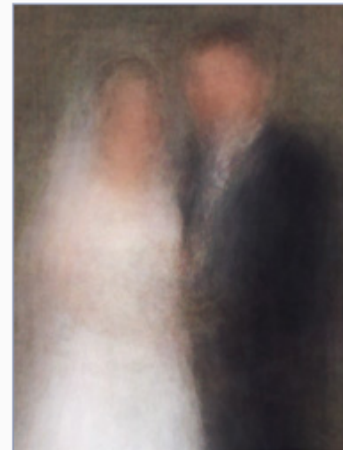
Little Leaguer



Kids with Santa



The Graduate



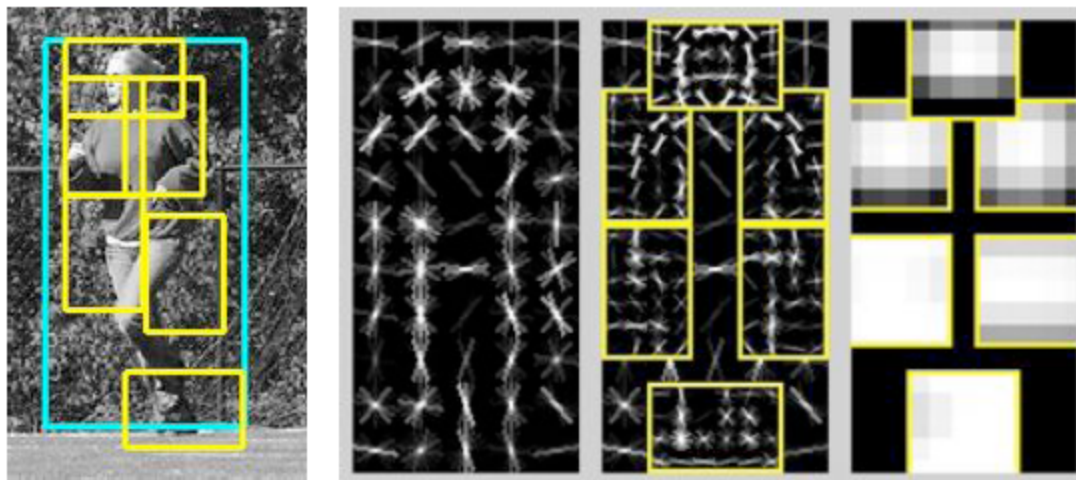
Newlyweds

Empirical Evaluation of Context
Divvala, Hoiem, Hays, Efros, Hebert,
CVPR 2009

Take a Good Detector and Make It Better

- UofCTTI from VOC 2007 (CVPR 2008)
- Many thanks to Pedro Felzenszwalb, David McAllester, and Deva Ramanan!

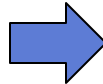
Deformable Parts Model



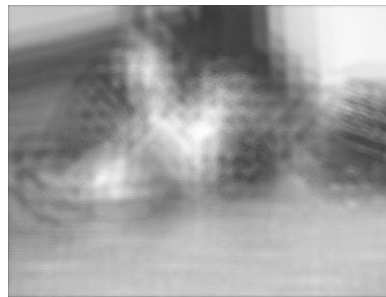
Likelihood of Object Presence

Image Statistics

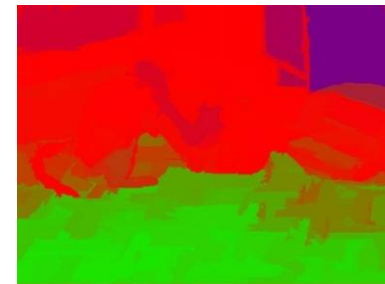
Image



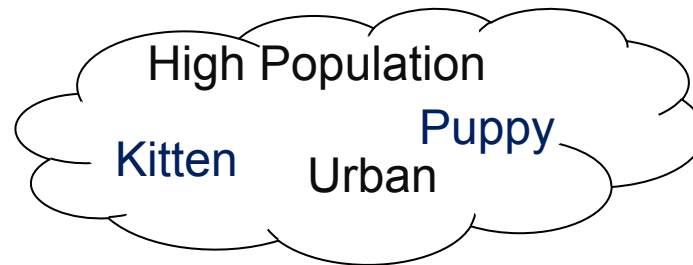
Gist



Geometric Context



Associated Data



gist: Torralba Oliva 2003
geom context: Hoiem et al. 2005
im2gps: Hays and Efros 2008

Context from Scene Match Properties

Scene Matches



...



Associated Tags/ Geo

Zoo ...
Bear

Urban ...
High Pop

Ruins ...
Maya

Forest ...
Low Pop

2007 ...
Riodejaneiro

Urban...
High Pop

Features for "Dining Table" Classifier

Keywords

Table: 0

Food: 0

Café: 1

Animals: 3

Nature: 4

Monument: 3

Hiking: 3

Verdict: Unlikely

Geography

Photo Density: 0.5

Pop Density: 0.6

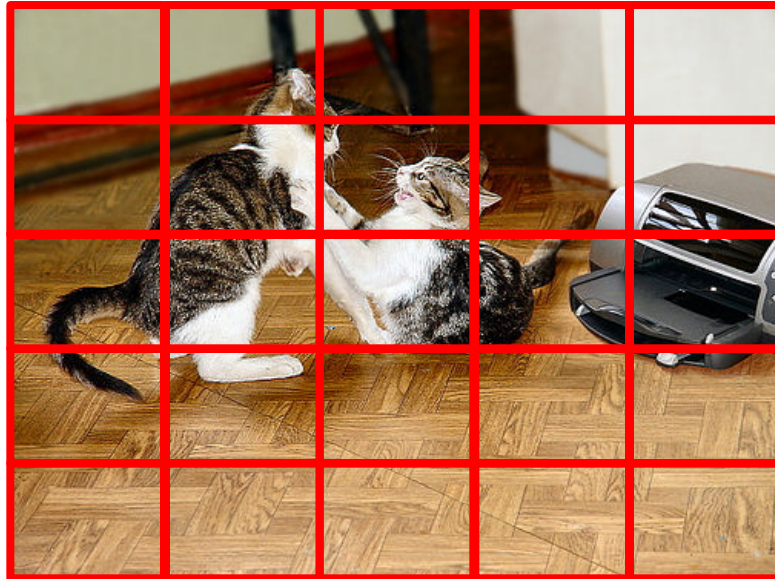
Forest: 0.4

RainForest: 0.2

Verdict: Not Sure

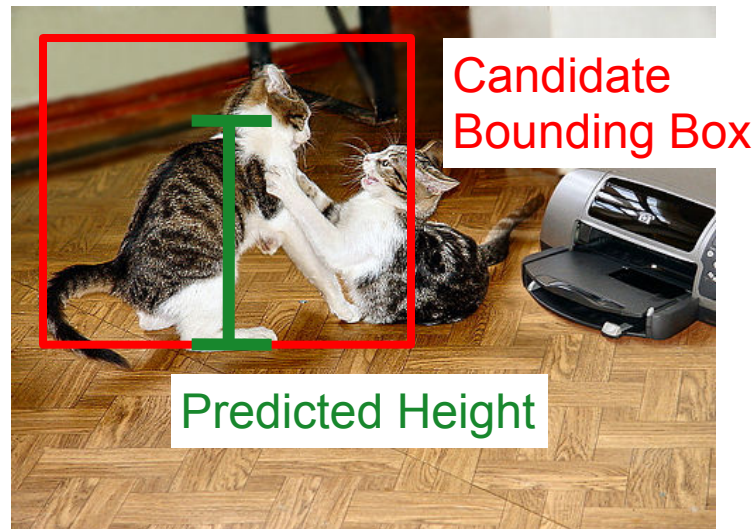
Likelihood of Object Location

- Build classifier for each cell based on whole image gist and geometric context



Likelihood of Object Size

- Predict bounding box height at given location
 - y-position
 - depth estimate at position
 - global gist and geometric context



Depth: Hoiem et al. 2007
Size from Gist: Torralba Oliva 2003

Score Combination

Independently Trained
Classifiers

Appearance Score
Window-Based Detector

Presence Scores
Gist + GC
Associated Data

Position Scores
Score in cell
Max in neighboring cells

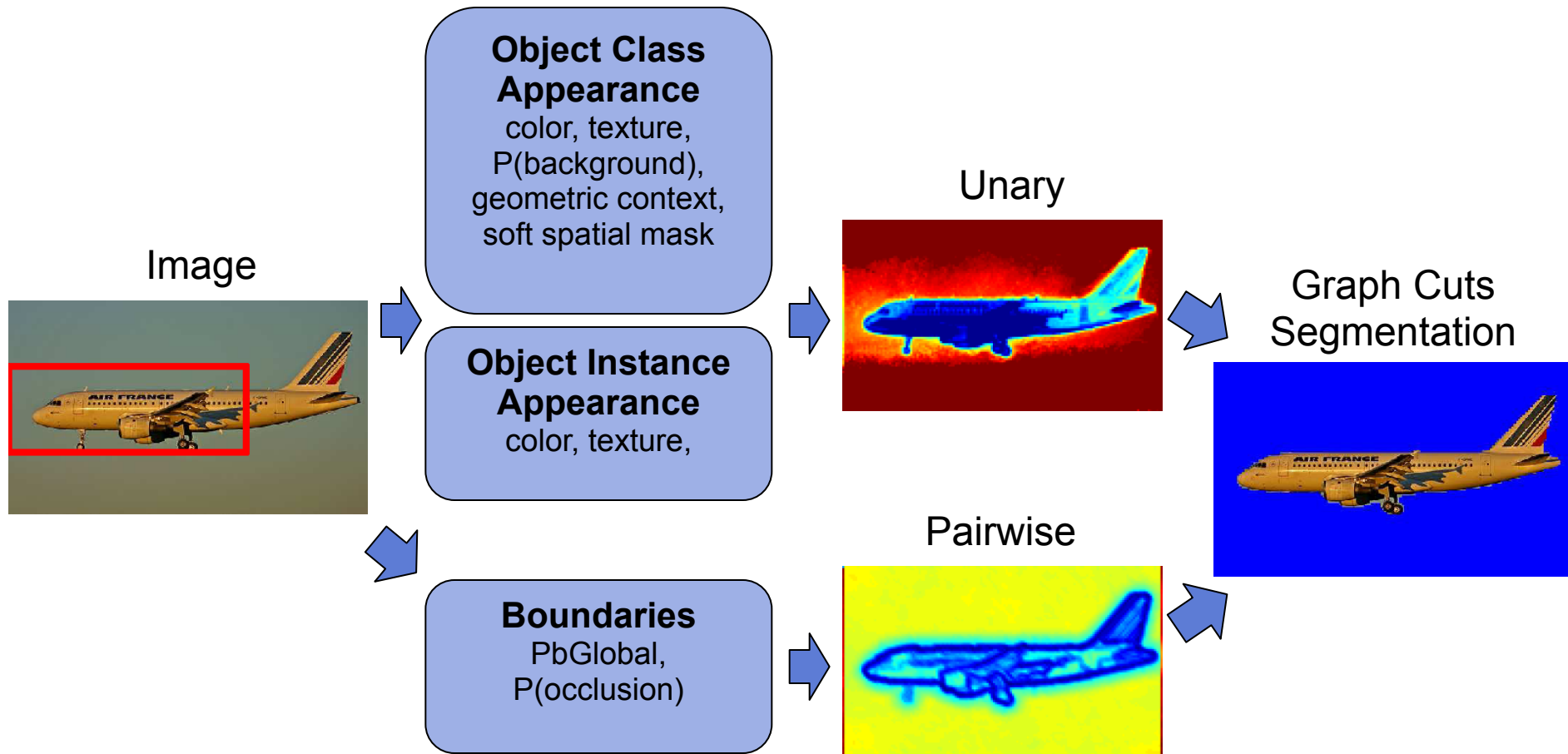
Size Scores
Box height
Diff from predicted height

Weights
L1-Regularized Logistic Regression



**Bounding Box
Score**

Object Support



PbGlobal: Maire et al. 2008

Occlusion: Hoiem et al. 2007

GraphCuts: Boykov et al. 2001

Analysis on PASCAL VOC 2008

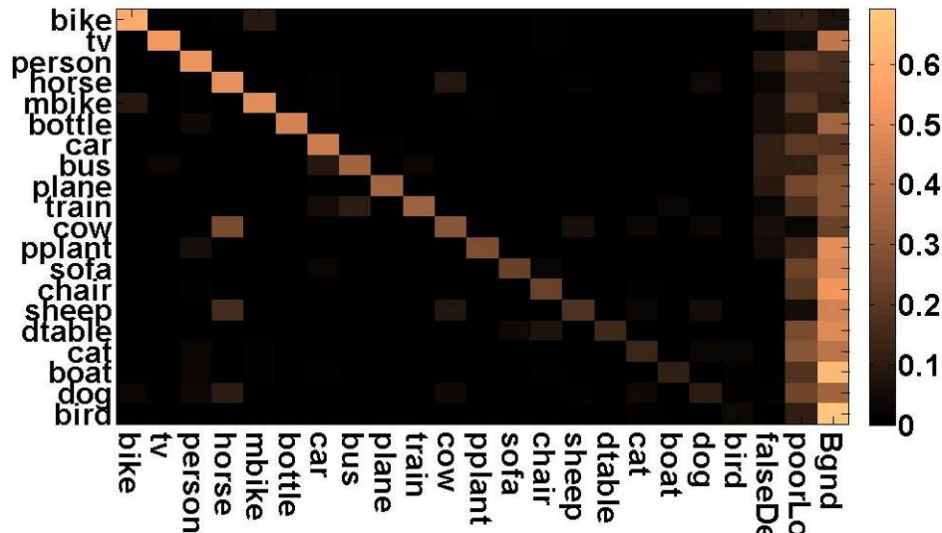
Overall Results on VOC 2008

Detector	Base	+Context	+Support
UoCTTI'07	18.2	19.4	22.0
UoCTTI'08	21.2	23.1	-

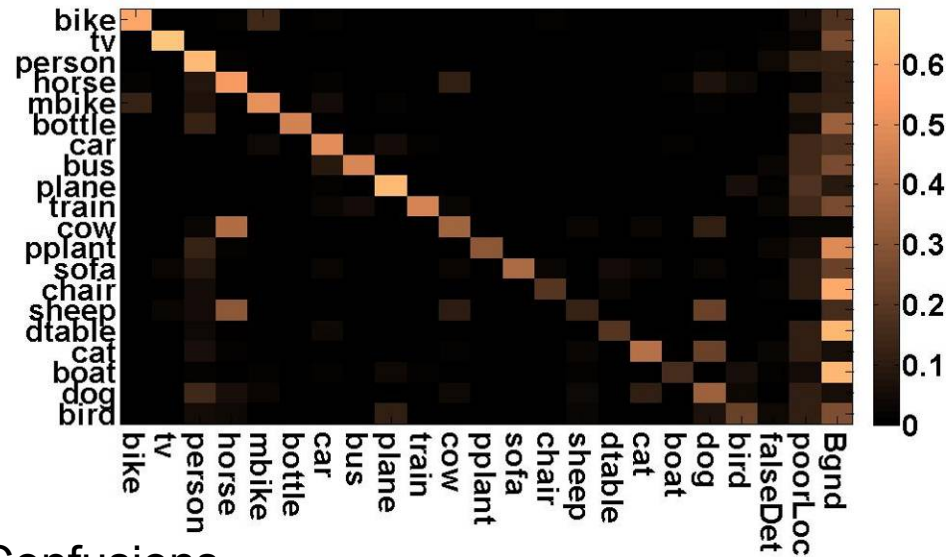
- Large Improvements: Airplanes, Cat, Dog, Person, Train
- Minor: Bottle, Potted plant, Chairs and Sheep

Confusion Matrices

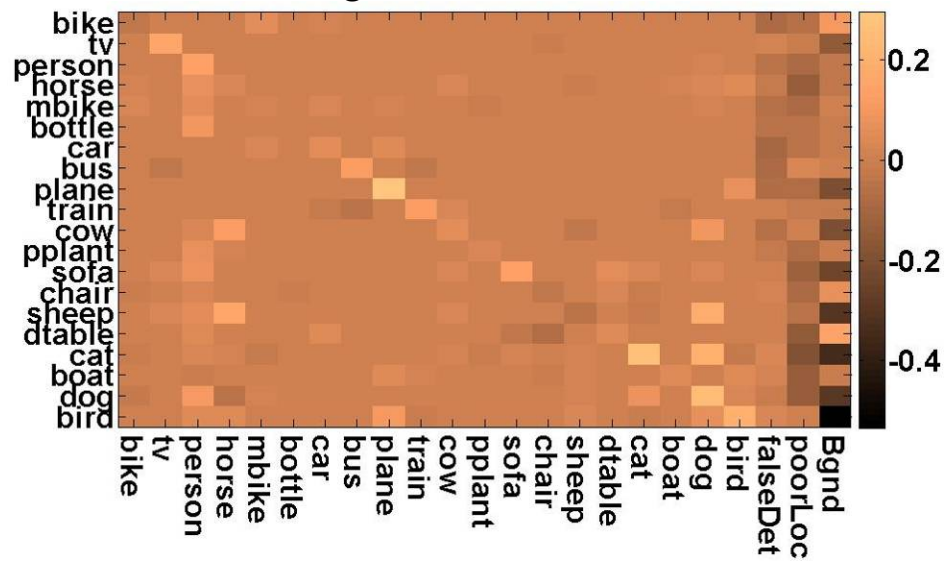
Without Context



With Context



Change in Confusions



Change in Accuracies

Type	Change
Small	97%
Large	15%
Occluded	79%
Non-Occluded	-5%
Difficult	57%
Total	47%

- Percentage change in average precisions before and after including context information
- Context helps in case of impoverished appearance

Context (feature) Importance

Context Type	Importance
Presence	3.2%
Semantic context	19.7%
Geographic context	9.2%
Location	16.5%
Size	51.5%

- Measured as normalized product of feature weight (assigned by the logistic regressor) and the mean variation of the feature

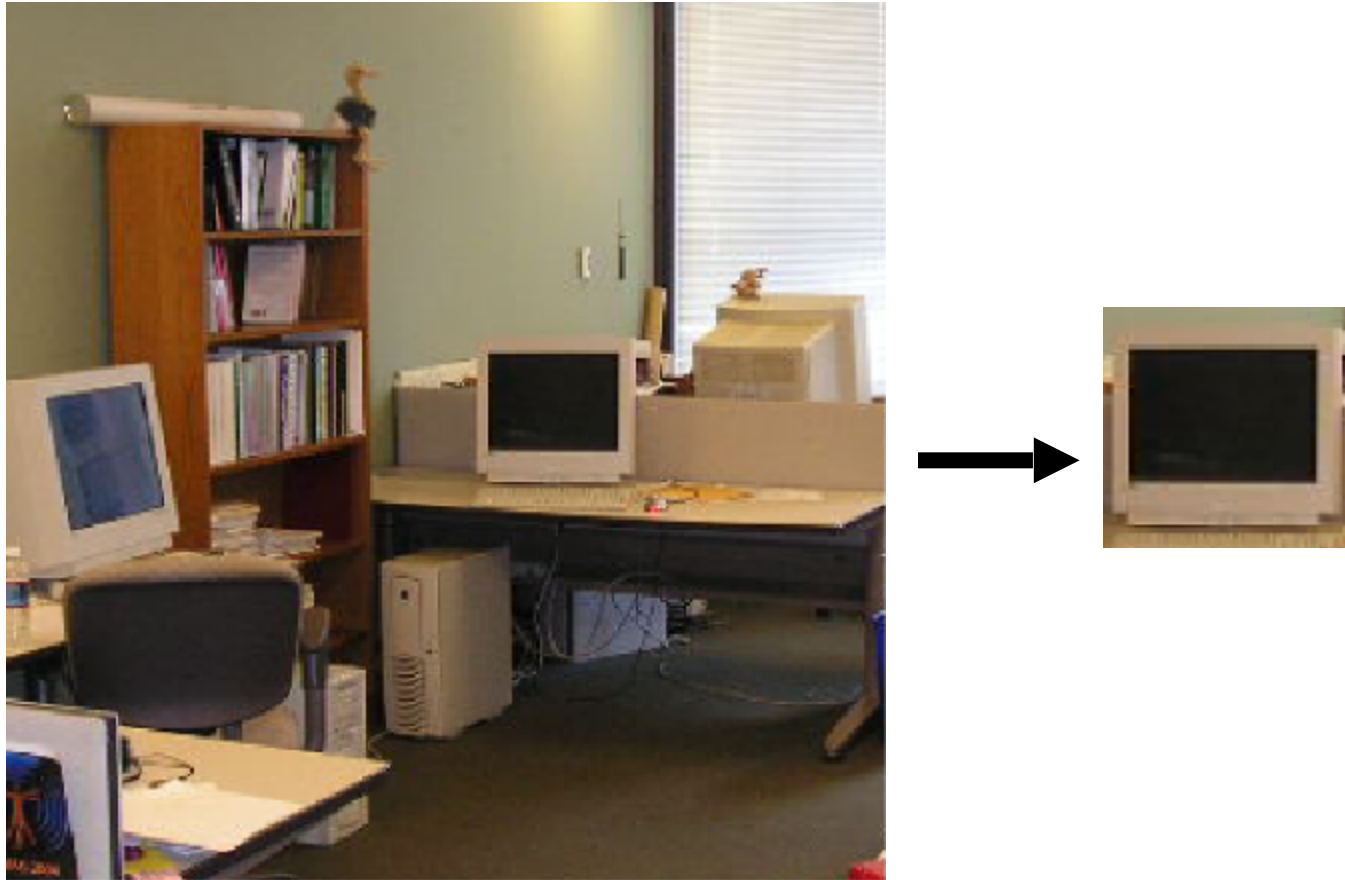
Detecting difficult objects



→ Office → Maybe there is a mouse

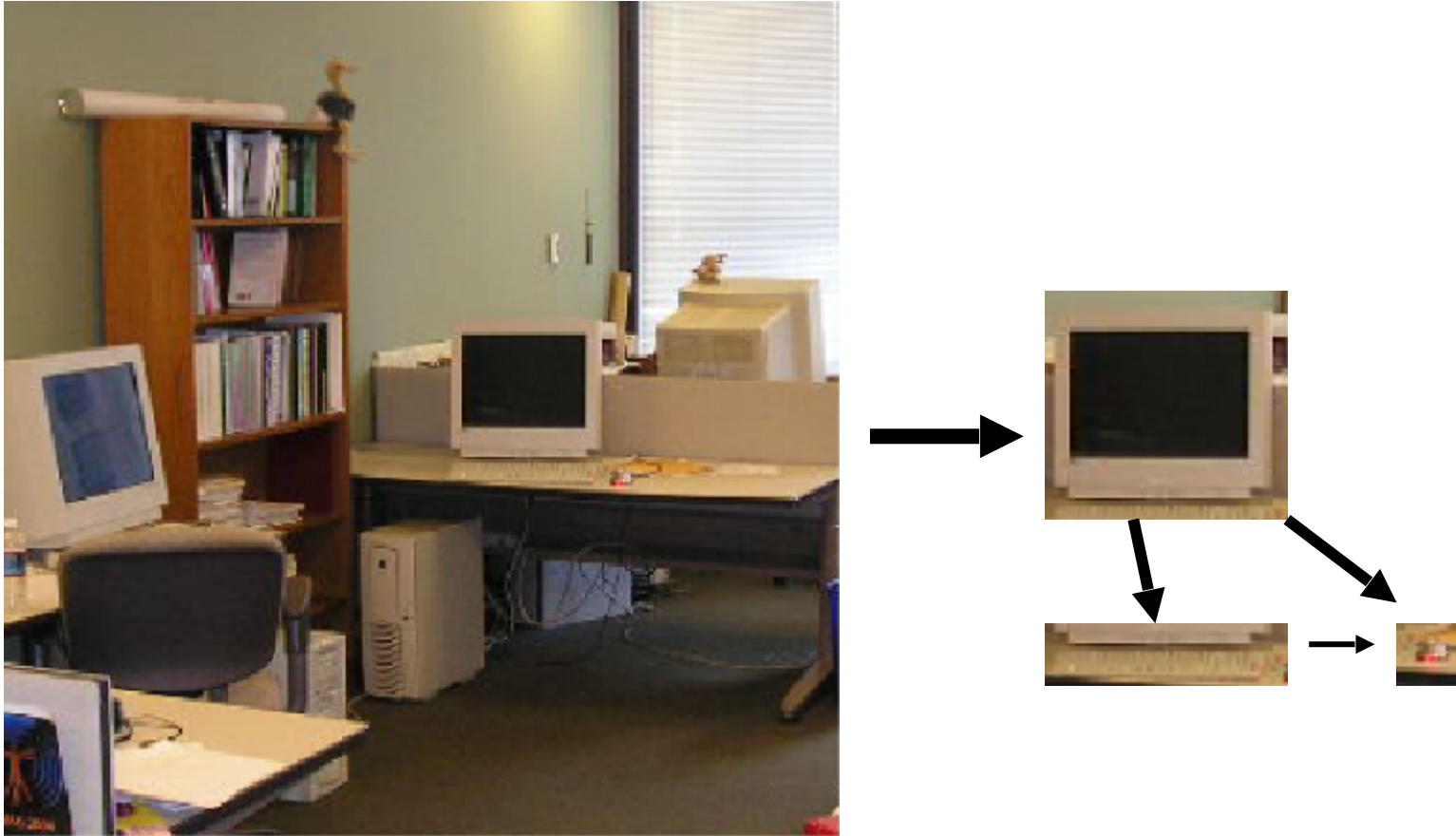
Start recognizing the scene

Detecting difficult objects



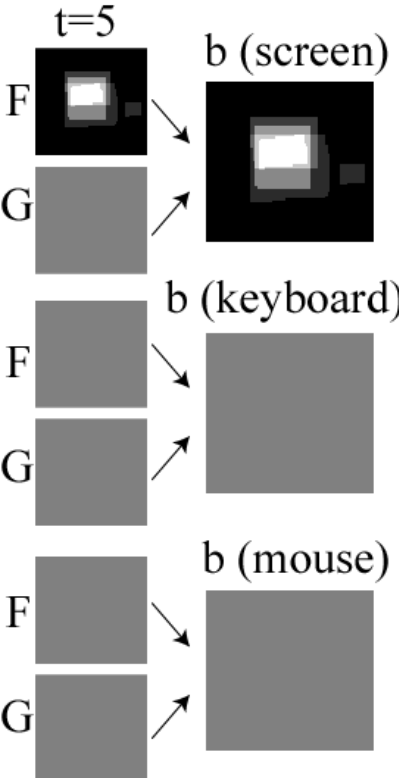
Detect first simple objects (reliable detectors) that provide strong contextual constraints to the target (screen -> keyboard -> mouse)

Detecting difficult objects (object parts)



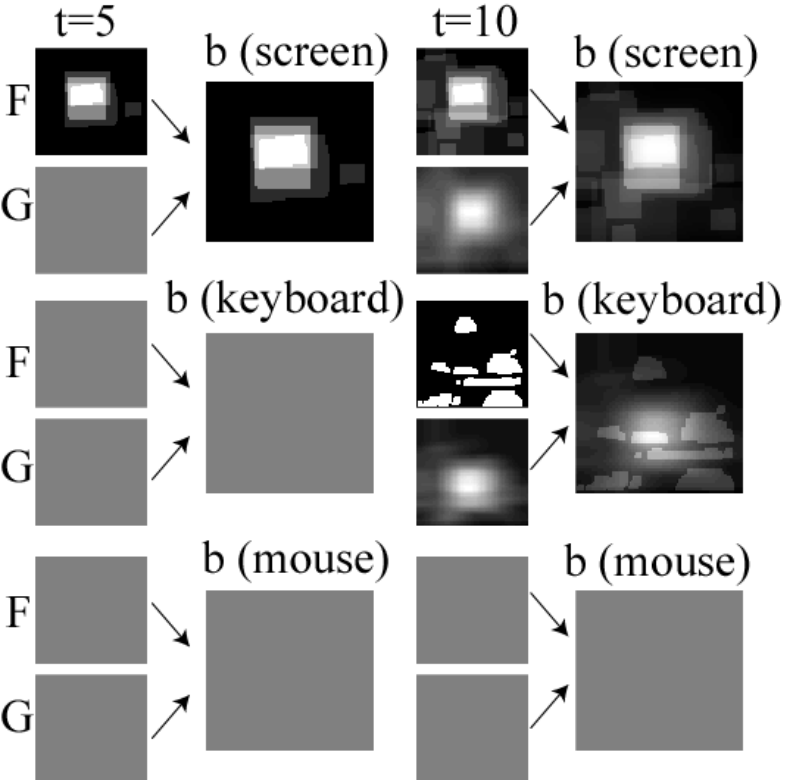
Detect first simple objects (reliable detectors) that provide strong contextual constraints to the target (screen $\rightarrow$ keyboard $\rightarrow$ mouse)

BRF for screen/keyboard/mouse



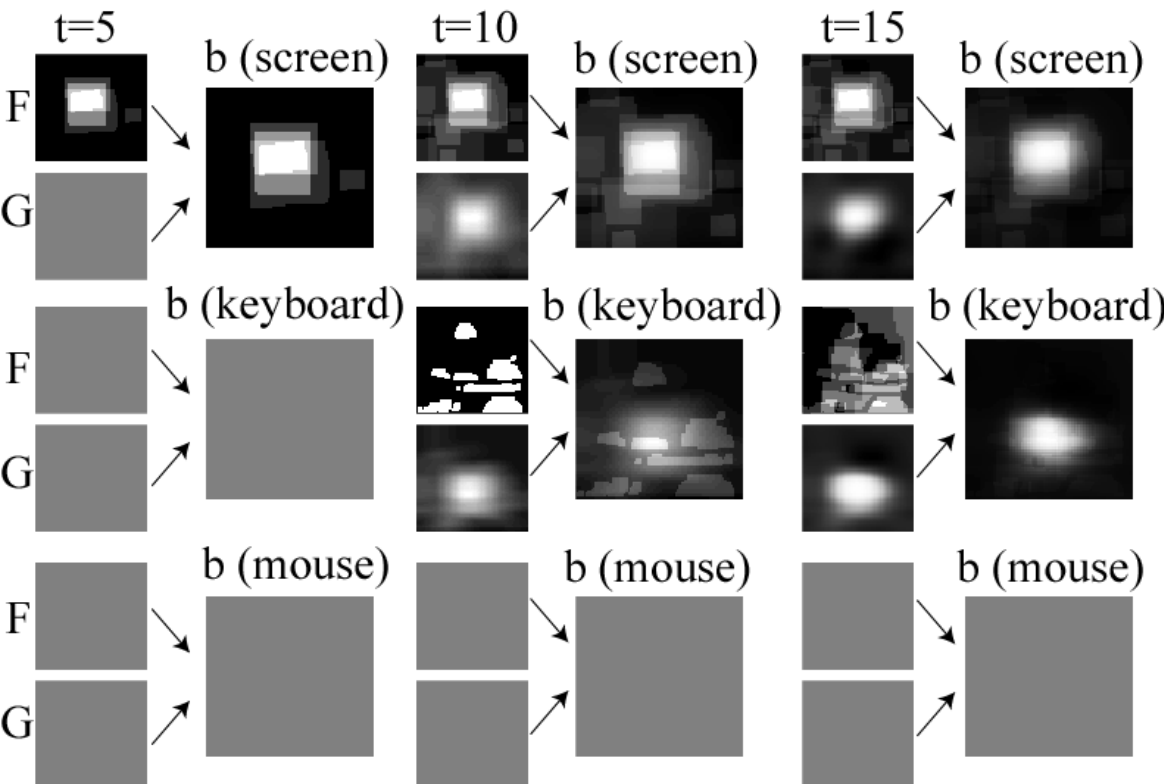
Iteration $\rightarrow$

BRF for screen/keyboard/mouse



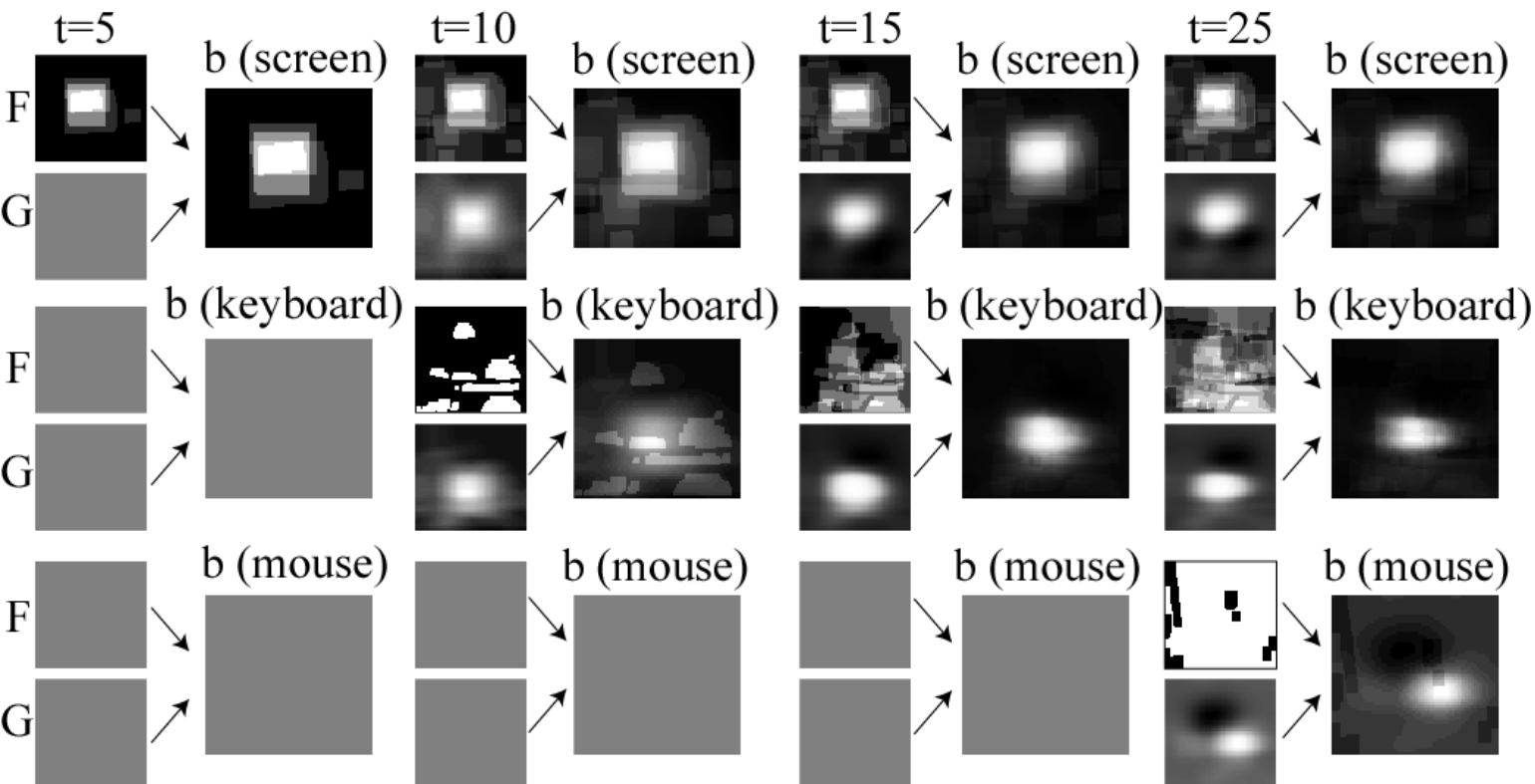
Iteration →

BRF for screen/keyboard/mouse



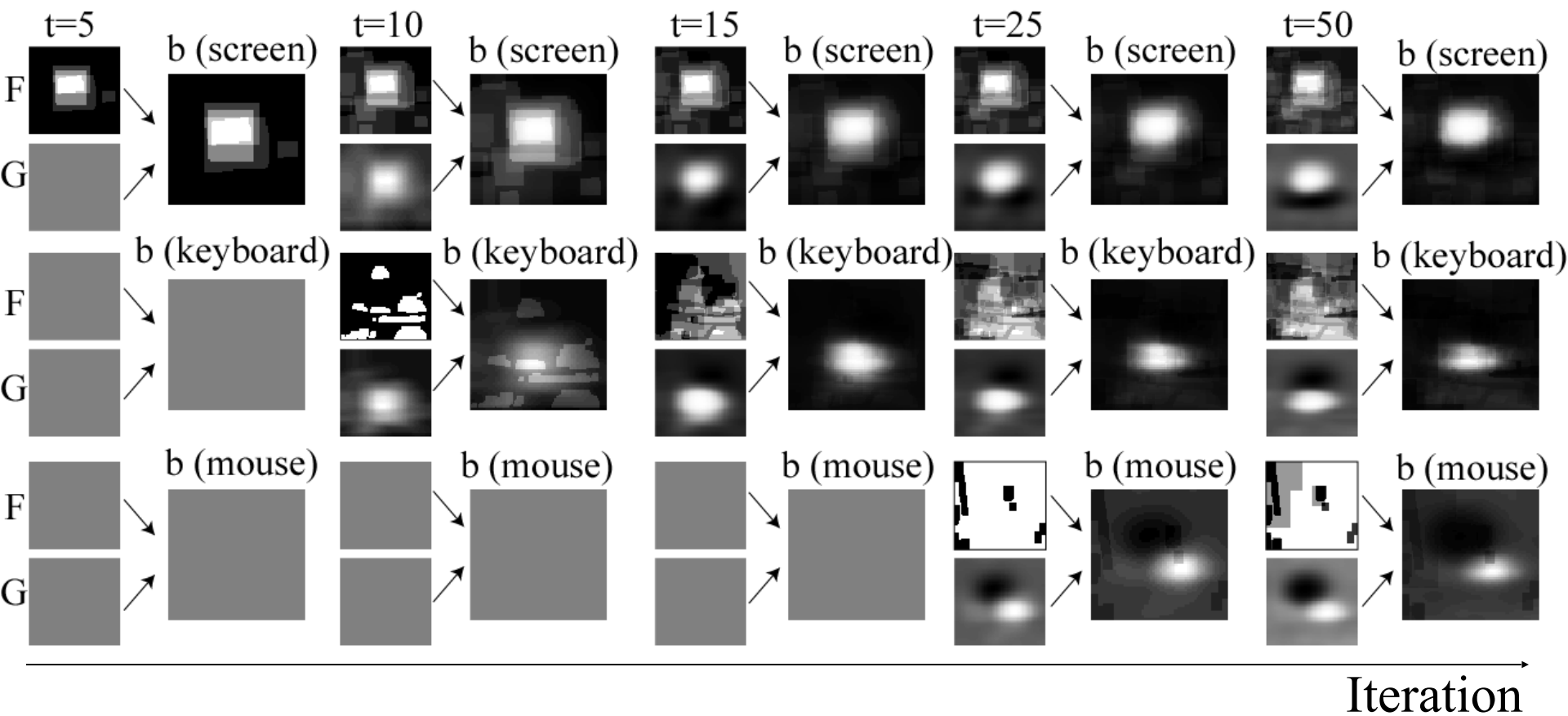
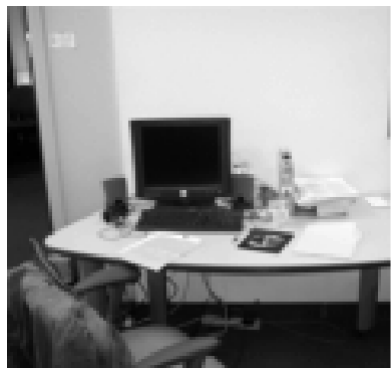
Iteration →

BRF for screen/keyboard/mouse

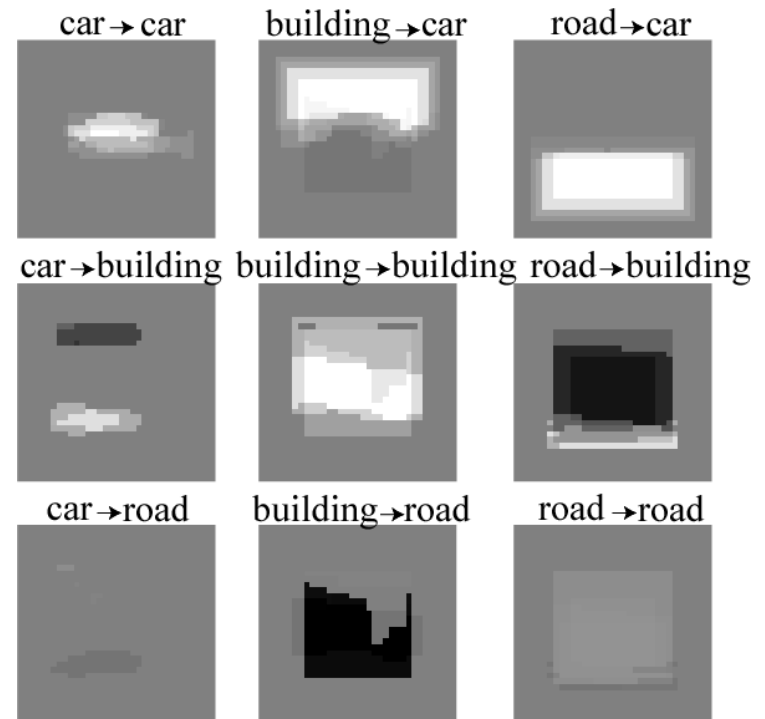
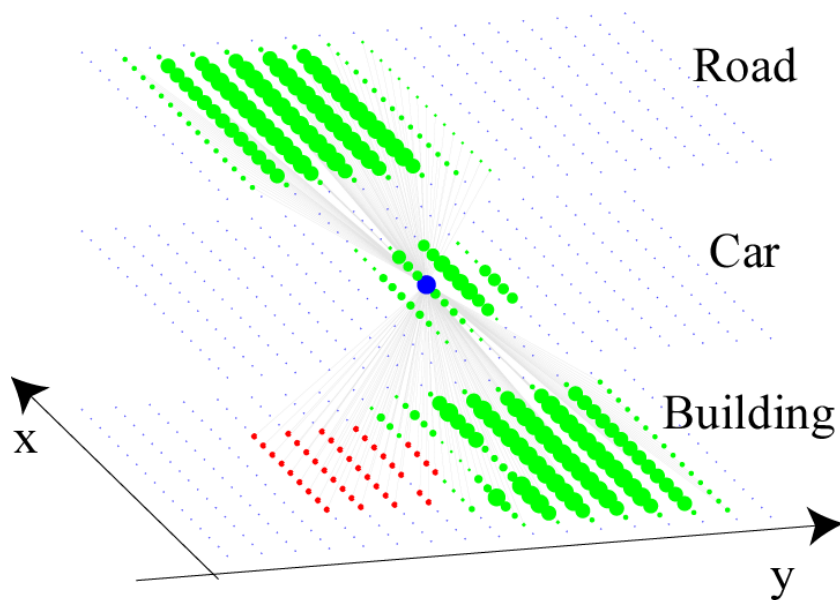


Iteration →

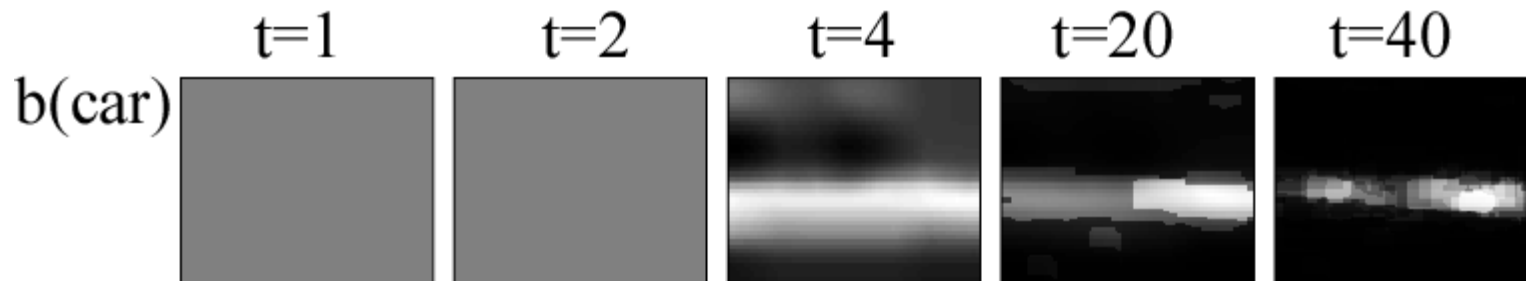
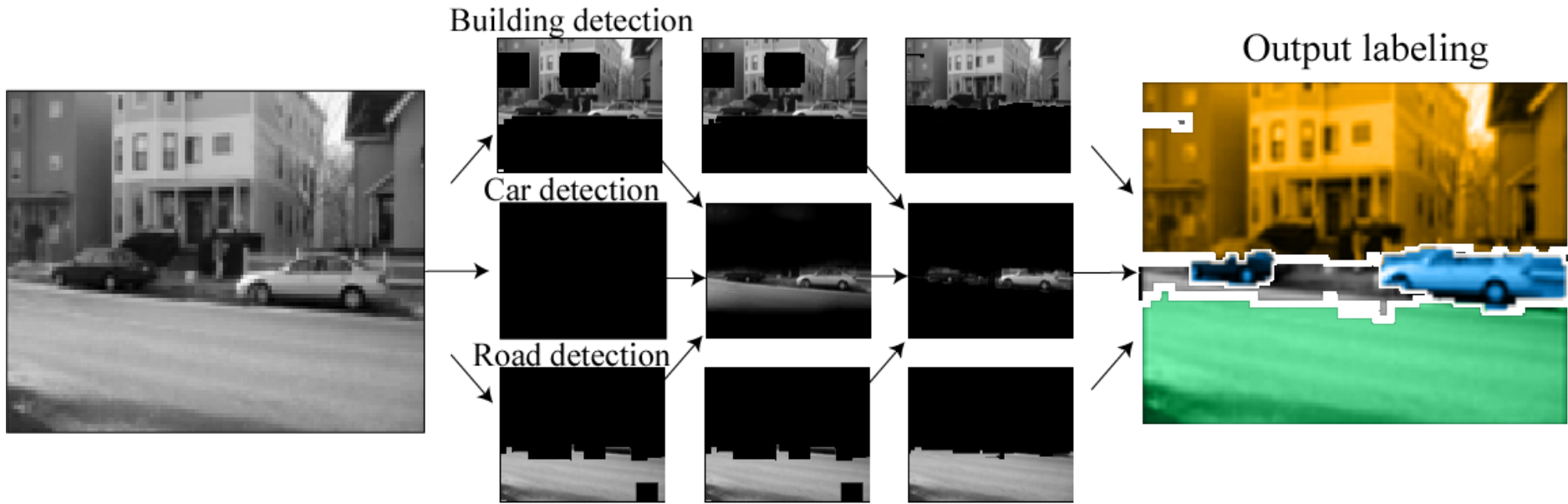
BRF for screen/keyboard/mouse



Object : context :: parts : objects

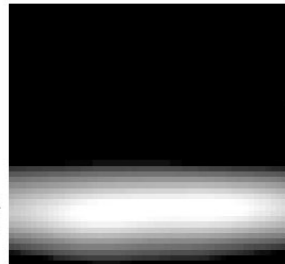
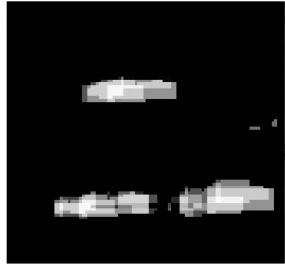
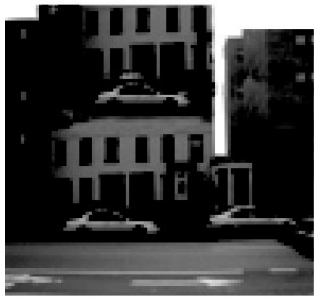


BRF for car detection: results

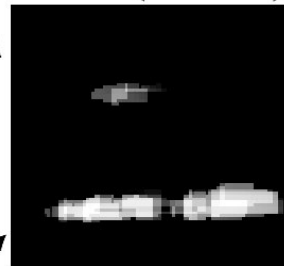


A “car” out of context is less of a car

From image F



$$b = \sigma(F + G)$$



Thresholded beliefs



From detectors G

Car

Building

Road



...on the other hand

Who needs context anyway?

We can recognize objects even out of context

